CDI COLLEGE

Assignment 2

Technical Analysis Using R – Preliminary Stage

Course	BDA400 – Data Science Tools and Techniques
Student	Mohamed Elahwel
Assignment	CA_BDA400_A02
Portfolio symbols	AAPL, MSFT, NVDA
Required filename	Elahwel_Mohamed_CA_BDA400_A02.pdf
Repository	https://github.com/_____/TechnicalAnalysis

Prepared as a submission-ready project package. Complete the highlighted evidence steps on your own computer and replace each screenshot placeholder with a genuine screenshot before submitting.

1. Introduction and Objectives

This preliminary stage establishes the technical environment for a three-stage stock technical-analysis project. The work covers GitHub repository setup, R and RStudio installation, package installation, portfolio configuration, stock-data importing, descriptive statistics, and output display.

The assignment objectives are to set up a GitHub account and repository; install R and RStudio; install relevant R packages; create utility functions to read stock data and compute statistics; initialize the technical-analysis process; display imported data and calculated statistics; and document the process.

2. GitHub Account and Repository

Create or sign in to a GitHub account, then create a repository named **TechnicalAnalysis**. Inside it, create a folder named **Assignment2** and place this report, portfolio.txt, and the R script in that folder. The assignment instructions require a repository link and a report documenting setup with screenshots.

Repository URL: https://github.com/_____/TechnicalAnalysis

Screenshot evidence to insert: GitHub profile/account page; TechnicalAnalysis repository page; Assignment2 folder showing the submitted files.

[SCREENSHOT PLACEHOLDER – Insert genuine GitHub account and repository screenshots here]

3. Installing and Configuring R and RStudio

Install R from CRAN, then install RStudio Desktop. Open RStudio and verify the installation by running the version commands below in the Console.

```
R.version.string\nRStudio.Version()$version
```

Screenshot evidence to insert: R installation/download or completion screen; RStudio installation/completion screen; RStudio open with the Console showing the version check.

[SCREENSHOT PLACEHOLDER – Insert genuine R and RStudio installation/verification screenshots here]

4. Installing Required R Packages

The script checks for and installs the packages quantmod, TTR, xts, and ggplot2. quantmod is used to retrieve stock market data, TTR supplies technical-analysis calculations such as the simple moving average, xts stores time-indexed market data, and ggplot2 creates price charts.

```
install.packages(c("quantmod", "TTR", "xts", "ggplot2"))\nlibrary(quantmod)\nlibrary(TTR)\nlibrary(xts)\nlibrary(ggplot2)
```

Screenshot evidence to insert: RStudio Console showing the package installation commands and successful package loading.

[SCREENSHOT PLACEHOLDER – Insert genuine package installation screenshots here]

5. Project Configuration and Initialization

Create a plain-text file named **portfolio.txt** in the same folder as the R script. It contains one ticker symbol per line. The portfolio for this version is:

```
AAPL\nMSFT\nNVDA
```

These are example U.S.-listed ticker symbols selected to demonstrate importing market data. The script requests approximately one year of daily historical data from Yahoo Finance through quantmod. Data availability depends on the symbol, internet connection, and the data provider.

6. Function to Import Stock Data

The **load_stock_data()** function reads portfolio.txt, removes blank lines, and loops through each ticker. It uses quantmod::getSymbols() with the Yahoo Finance source and stores each successful result in a named list. A warning is generated for a symbol that cannot be downloaded, rather than stopping the entire import process.

The downloaded objects are time-series data (xts) containing Open, High, Low, Close, Volume, and Adjusted price fields when supplied by the data source. Each symbol is stored separately in portfolio_data.

7. Function to Compute Basic Statistics

The **calculate_statistics()** function extracts closing prices and calculates: the 20-day simple moving average (latest available value), mean, statistical mode, median, and standard deviation. The mode is calculated with a custom frequency function because base R's mode() returns an object's storage type rather than its most frequent value.

The statistics are returned as a structured data frame for each stock. If there are no valid closing prices, or the expected Close column is unavailable, the function reports an error.

8. Displaying Data and Visualizations

The **display_stock_data()** function prints the most recent six rows for each symbol and its summary statistics. The **plot_stock_data()** function draws a closing-price time-series line chart for each stock. A combined statistics table is also printed and saved as stock_statistics.csv.

Screenshot evidence to insert after running the script: RStudio Console showing imported rows and statistics; one or more generated closing-price charts; the saved CSV file in the Files pane.

[SCREENSHOT PLACEHOLDER – Insert genuine R output, statistics, charts, and CSV evidence here]

9. How to Run the Project

1. Download or clone the TechnicalAnalysis repository and open the Assignment2 folder in RStudio. 2. Confirm portfolio.txt and the R script are in the working directory. 3. Open the R script and click Source, or run the code from top to bottom. 4. Allow package installation if prompted. 5. Check the Console for downloaded data and statistics. 6. Check the Plots pane for charts and the Files pane for stock_statistics.csv. 7. Save all files in the Assignment2 folder and push/commit them to GitHub.

10. Files Included in This Package

• This PDF report. • portfolio.txt with the selected stock symbols. • technical_analysis.R with package setup, import, statistics, display, visualization, and CSV export functions. • README.txt with run and submission notes.

11. Conclusion

This preliminary project provides a reusable R workflow for reading a portfolio of stock symbols, importing historical stock prices, calculating the required basic statistics, and displaying the data and charts. The repository and evidence screenshots should be finalized on the student's computer before submission.

12. Final Submission Checklist

Item	Status before submission
GitHub repository named TechnicalAnalysis	Create/verify and add repository link
Assignment2 subfolder	Create and upload all project files
R and RStudio installed	Verify on your PC and add screenshots
quantmod, TTR, xts, ggplot2 installed	Run script and add Console screenshot
portfolio.txt	Included in package
technical_analysis.R	Included in package; run and verify
Data, statistics, chart output	Run on your PC and add genuine screenshots
Report saved with college filename	PDF filename provided

Important academic-integrity note: the screenshot areas are placeholders, not evidence of completed installation or repository activity. Capture your own screenshots after performing the steps. Do not submit this report with the placeholders if your instructor requires screenshots.